

ITSOLERA Pvt. Ltd.

Machine Learning Internship Project Proposal

Project Title: Time-Series Demand Forecasting System with External Variables

1. Introduction

Accurate demand forecasting is critical for businesses in retail, manufacturing, and supply chain management. Poor forecasting can lead to overstocking, stockouts, increased costs, and lost revenue.

This project focuses on developing a **Time-Series Demand Forecasting System** that not only uses historical sales data but also incorporates **external variables** such as promotions, holidays, weather conditions, and economic indicators. Unlike simple forecasting models, this system integrates multiple influencing factors to improve prediction accuracy.

The project emphasizes **time-series modeling, feature engineering for temporal data, external variable integration, and advanced evaluation techniques.**

2. Objectives

The objectives of this project are:

- To build a time-series forecasting model for product demand
- To incorporate external influencing variables into forecasting
- To compare classical statistical models and machine learning approaches
- To evaluate forecasting accuracy using time-series metrics
- To generate actionable insights for inventory and supply planning

3. Literature Review

Time-series forecasting has been widely used in retail analytics and supply chain optimization. Traditional statistical models such as ARIMA and SARIMA are effective for modeling seasonality and trends. However, they often struggle to incorporate multiple external variables.

Recent studies show that machine learning models such as Random Forest, Gradient Boosting, and Prophet can better handle external regressors and nonlinear relationships. Including external factors such as holidays and promotions significantly improves forecasting performance compared to using historical demand alone.

4. Methodology

4.1 Data Collection

The dataset will include:

Date
Product ID
Daily / Weekly Sales Quantity
Promotion Indicator
Holiday Indicator
Weather Data (Temperature, Rainfall)
Economic Indicators (Optional)

Suggested publicly available datasets:

Retail Sales Forecasting Dataset (Kaggle)
Walmart Sales Forecasting Dataset (Kaggle)
Rossmann Store Sales Dataset (Kaggle)
Weather Dataset (NOAA or Kaggle)

4.2 Data Preprocessing

Convert date column into datetime format

Extract time-based features:

Day
Week
Month
Quarter
Year
Day of Week

Handle missing values

Detect and treat outliers

Normalize or scale features if required

Split dataset using time-based splitting (not random splitting)

5. Tools & Technologies

Programming Language: Python

Libraries:

Pandas, NumPy
Statsmodels (ARIMA / SARIMA)
Facebook Prophet

Scikit-learn
XGBoost

Time-Series Techniques:
ARIMA / SARIMA
Prophet with external regressors
Machine Learning Regression Models

Evaluation Metrics:
MAE (Mean Absolute Error)
RMSE (Root Mean Squared Error)
MAPE (Mean Absolute Percentage Error)

Visualization:
Matplotlib, Seaborn

Development Environment:
Jupyter Notebook / VS Code

Version Control:
Git & GitHub

4.3 Model Development

Develop and compare multiple forecasting approaches:

Baseline Model (Naïve Forecast)

ARIMA / SARIMA model

Prophet model with external regressors

Machine Learning models such as:

Random Forest Regressor
XGBoost Regressor

Incorporate external variables into advanced models and compare performance.

4.4 Model Evaluation

Evaluate forecasting models using:

MAE
RMSE
MAPE

Perform rolling-window validation for realistic forecasting evaluation.

Visualize predicted vs actual demand trends.

Analyze performance during high-demand seasons and promotional periods.

6. Expected Outcomes

A robust demand forecasting system

Improved forecasting accuracy using external variables

Comparison of statistical and machine learning time-series models

Better understanding of temporal feature engineering

Insights for inventory management and supply planning

7. Conclusion

This project aims to develop a comprehensive time-series demand forecasting system enhanced with external influencing variables. By integrating historical sales data with contextual factors such as promotions and holidays, the system provides more accurate and reliable forecasts. The project offers hands-on experience in time-series modeling, feature engineering, advanced evaluation methods, and real-world business forecasting applications.

